

2 Description

The STM32C031x4/x6 mainstream microcontrollers are based on high-performance Arm® Cortex®-M0+ 32-bit RISC core operating at up to 48 MHz frequency. Offering a high level of integration, they are suitable for a wide range of applications in consumer, industrial and appliance domains and ready for the Internet of Things (IoT) solutions.

The devices incorporate a memory protection unit (MPU), high-speed embedded memories (12 Kbytes of SRAM and up to 32 Kbytes of flash program memory with read and write protection), DMA, an extensive range of system functions, enhanced I/Os, and peripherals. The devices offer standard communication interfaces (one I²Cs, one SPI / one I²S, and two USARTs), one 12-bit ADC (2.5 MSps) with up to 21 channels, a low-power RTC, an advanced control PWM timer, four general-purpose 16-bit timers, two watchdog timers, and a SysTick timer.

The devices operate within ambient temperatures from -40 to 125°C and with supply voltages from 2.0 V to 3.6 V. Optimized dynamic consumption combined with power-saving modes allows the design of low-power applications.

The devices are housed in packages with 20 to 48 pins.

Table 2. STM32C031x4/x6 family device features and peripheral counts

Peripheral		STM32C031_							
		_F4	_F6	_G4	_G6	_K4	_K6	_C4	_C6
Flash memory (Kbyte)		16	32	16	32	16	32	16	32
SRAM (Kbyte)		12 with parity							
Timers	Advanced control	1 (16-bit)							
	General-purpose	4 (16-bit)							
	SysTick	1							
	Watchdog	2							
Comm. interfaces	SPI [I2S] ⁽¹⁾	1 [1] + 2 extra through USARTs							
	I2C	1							
	USART	2							
RTC / RNG / AES / VREFBUF		Yes / No / No / No							
GPIOs (all 5V-tolerant)		18		26		30		45	
DMA channels		3							
Wake-up pins		4				5			
12-bit ADC channels (ext. + int.)		13 + 2		15 + 2		16 + 2		19 + 2	
Max. CPU frequency		48 MHz							
Operating voltage		2.0 to 3.6 V							
Operating temperature ⁽²⁾		Ambient: -40 to 85 °C / -40 to 105 °C / -40 to 125 °C Junction: -40 to 105 °C / -40 to 125 °C / -40 to 130 °C							